

Article

The bottleneck of diffusion and inductive effects in LiGeSnPS

Thorben Krauskopf, Sean P. Culver, and Wolfgang G. Zeier

Chem. Mater., **Just Accepted Manuscript** • DOI: 10.1021/acs.chemmater.8b00266 • Publication Date (Web): 16 Feb 2018Downloaded from <http://pubs.acs.org> on February 17, 2018**Just Accepted**

"Just Accepted" manuscripts have been peer-reviewed and accepted for publication. They are posted online prior to technical editing, formatting for publication and author proofing. The American Chemical Society provides "Just Accepted" as a service to the research community to expedite the dissemination of scientific material as soon as possible after acceptance. "Just Accepted" manuscripts appear in full in PDF format accompanied by an HTML abstract. "Just Accepted" manuscripts have been fully peer reviewed, but should not be considered the official version of record. They are citable by the Digital Object Identifier (DOI®). "Just Accepted" is an optional service offered to authors. Therefore, the "Just Accepted" Web site may not include all articles that will be published in the journal. After a manuscript is technically edited and formatted, it will be removed from the "Just Accepted" Web site and published as an ASAP article. Note that technical editing may introduce minor changes to the manuscript text and/or graphics which could affect content, and all legal disclaimers and ethical guidelines that apply to the journal pertain. ACS cannot be held responsible for errors or consequences arising from the use of information contained in these "Just Accepted" manuscripts.

**ACS Publications**

The bottleneck of diffusion and inductive effects in $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$

Thorben Krauskopf^a, Sean P. Culver^{a,b}, Wolfgang G. Zeier^{*a,b}

^a*Institute of Physical Chemistry, Justus-Liebig-University Giessen, Heinrich-Buff-Ring 17, D-35392 Giessen, Germany.*

^b*Center for Materials Research (LaMa), Justus-Liebig-University Giessen, Heinrich-Buff-Ring 16, D-35392 Giessen, Germany.*

Abstract

The lithium ion conductor $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (LGPS) is known to exhibit ionic conductivity values as high as $12 \text{ mS}\cdot\text{cm}^{-1}$. Unfortunately, counter to chemical intuition, many attempts to enhance the ionic transport in LGPS, e.g. by increasing the Sn-fraction in $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$, have even led to a reduction in the conductivity. Employing a combination of Rietveld refinements against X-ray diffraction data, speed of sound measurements and electrochemical impedance spectroscopy, we investigate the structure-property relationships governing this behavior. Herein, it is shown that with increasing Sn^{4+} fraction in $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$, a structural bottleneck along the diffusion channels in the z -direction begins to tighten and with the concomitant increase in the lattice softness, the local ionic bonding interactions between Li^+ and S^{2-} become stronger, further increasing the activation barrier. This work provides a likely explanation for the lower conductivity exhibited by $\text{Li}_{10}\text{SnP}_2\text{S}_{12}$ and demonstrates that there is more to the underlying lithium diffusion mechanism in the $\text{Li}_{10}\text{MP}_2\text{S}_{12}$ structure.

1. Introduction.

Lithium thiophosphates are currently attracting interest for use in solid-state batteries, as a replacement for the currently commercial liquid electrolytes.^{1–3} In addition to compounds such as the argyrodites $\text{Li}_6\text{PS}_5\text{X}$,^{4–8} $\text{Li}_2\text{S-P}_2\text{S}_5$ glasses and the glass-ceramics,^{9–15} the most exciting solid electrolyte has indeed been $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (LGPS), since its initial discovery in 2011.¹⁶ Despite work already incorporating LGPS into the assembly of solid-state batteries,^{17–19} there are still a number of unanswered questions regarding the underlying structural and transport principles governing the outstanding properties in this material.

Typical approaches toward gaining a better understanding of the structure-property relationships in LGPS have involved the isoelectronic substitution of Ge with Si or Sn,^{20–24} or even the modulation of charge carrier densities via aliovalent substitution in $\text{Li}_{10+x}\text{M}_{1-x}\text{P}_{2-x}\text{S}_{12}$ ($\text{M} = \text{Si, Ge, Sn}$).^{21,25–27} Figure 1a,b shows the crystal structure of $\text{Li}_{10}\text{MP}_2\text{S}_{12}$ with its immobile PS_4 and $(\text{M/P})\text{S}_4$ tetrahedra. Notably, there are distinct tunnels hosting Li1 and Li3 along the z-axis that serve as fast transport channels (see Figure 1c), while the Li2 and Li4 positions connect to Li1 and Li3 in the x-y plane. Initially, LGPS was thought to exhibit purely one-dimensional lithium transport,¹⁶ with three-dimensional diffusion only being achieved at elevated temperatures,²⁷ despite theoretical considerations showing that a three-dimensional conduction mechanism might be at play.^{21,28} More recently, nuclear magnetic resonance and neutron diffraction further suggested that lithium diffusion also occurs in the x-y plane, in addition to the tunnel-like diffusion along the z-axis.^{29,30}

However, if $\text{Li}_{10}\text{MP}_2\text{S}_{12}$ is in fact a three-dimensional conductor exhibiting isotropic lithium diffusion, then it is expected that the conductivity should increase with increasing lattice volume. Contrary to this, ab initio molecular dynamics simulations²¹ predicted that an expansion of the LGPS unit cell (e.g. in $\text{Li}_{10}\text{SnP}_2\text{S}_{12}$) would not significantly increase the Li^+ conductivity. A prediction that was later confirmed experimentally.²⁴ Moreover, it was also proposed that a decrease in the lattice volume would cause a drastic reduction of the conductivity, thereby suggesting that LGPS itself possesses the optimal structural framework for Li^+ diffusion.²¹ In order to obtain a better understanding of the optimum channel size, Kato *et al.*²³ investigated a series of $\text{Li}_{10}\text{Ge}_{1-x}\text{M}_x\text{P}_2\text{S}_{12}$ ($\text{M} = \text{Si, Sn}$) solid solutions and found that the composition with the maximum conductivity and the lowest activation barrier was close to the stoichiometry of $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$. Nevertheless, while the smaller Si leads to a lower conductivity due to restricted diffusion pathways,²⁰ the structural reasons behind the decrease

in conductivity and the associated increase in the activation barrier upon moving from Ge^{4+} to Sn^{4+} are still unresolved.

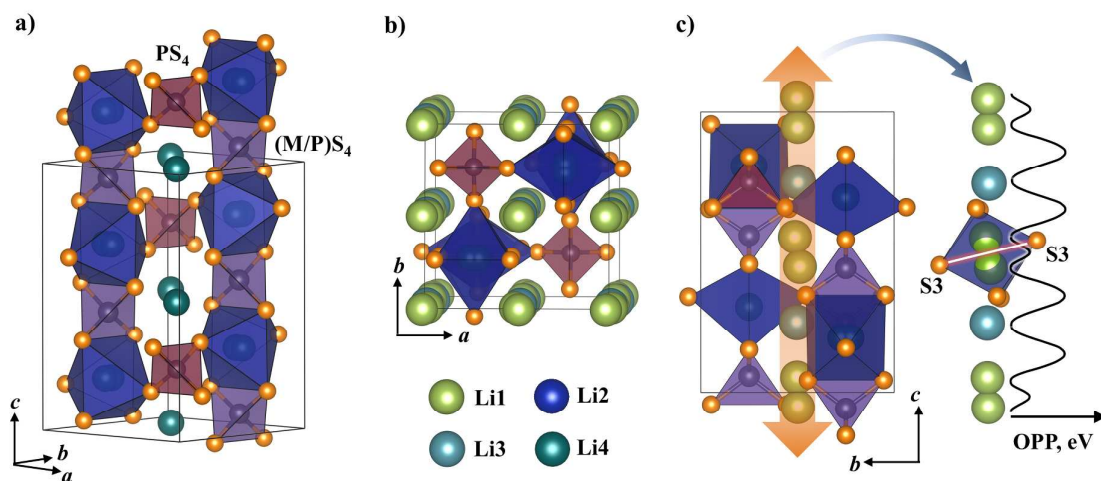


Figure 1: a) and b) Crystal structure of $\text{Li}_{10}\text{MP}_2\text{S}_{12}$ showing the PS_4 and $(\text{M/P})\text{S}_4$ tetrahedra of the framework, as well as the tunnels of Li1 and Li3 along the z -axis. c) Li1 and Li3 distribution for the predominant conduction pathway in the $\langle 001 \rangle$ -direction and projected one-particle potentials (OPP).²⁹ The Li1-Li1 jump is connected via an apical S3-S3 distance, showing that the local bonding environment is crucial for complex ion transport mechanisms.

Inspired by the remaining question as to why increasing the lattice volume of LGPS leads to a reduction in the conductivity and an increase in the activation barrier, we investigate the structural and transport changes in $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$ ($x = 0, 0.33, 0.67$ and 1). By combining Rietveld refinements against X-ray diffraction data, speed of sound measurements and electrochemical impedance spectroscopy, a plausible explanation has been elucidated. While increasing the Sn content increases both the lattice volume and the c/a ratio, the pathway along the z -direction becomes narrower and the local bonding interactions between Li^+ and S^{2-} become stronger, both of which increase the activation barrier and decrease the conductivity. This work demonstrates that while a lattice expansion is often expected to increase the conductivity, it can also result in local contractions that induce an additional bottleneck for diffusion and actually hinder ionic conduction.

2. Experimental Methods

Synthesis. All preparations and sample treatments were carried out under argon atmosphere ($O_2 < 1$ ppm, $H_2O < 5$ ppm). $Li_{10}Ge_{1-x}Sn_xP_2S_{12}$ was synthesized using the following procedure: the starting materials of lithium sulfide (Li_2S , Sigma Aldrich, 99.98%), phosphorus pentasulfide (P_2S_5 , Sigma-Aldrich, 99%), elemental sulfur (S_8 , Arcos Organics, >99.999%), germanium sulfide (GeS , Sigma-Aldrich, 99.99%) and tin sulfide (SnS , Sigma-Aldrich, > 99.99%) were mixed in the appropriate stoichiometric ratio. Additionally, a 3 wt% excess of sulfur was added to the mixture to compensate for sulfur loss at higher temperatures. The resultant mixture (3 g) was ball milled (Fritsch Pulveristette 7 premium line) at 400 rpm using a ZrO_2 milling set (80 mL bowl with 90 g of 3 mm diameter balls). The milling was performed for 48 h with intermediate cooling times (i.e. 15 min of cooling after every 10 min of milling) to prevent excessive heating of the samples. Twice during the process, the grinding bowl was opened and the resultant mixture was ground to obtain a uniform precursor. The resultant precursor (1g) was pressed into pellets, which were then sealed under vacuum into 10 mm inner-diameter quartz ampoules. The ampoules were heated in a tube furnace (Nabertherm) to 773 K (at $27\text{ K}\cdot\text{h}^{-1}$), annealed for 20 h and then cooled to room temperature. In order to reduce side phases in $Li_{10}SnP_2S_{12}$, the pellet was reheated at 873 K for 48 h (at $50\text{ K}\cdot\text{h}^{-1}$). The powder was then ground for 10 min and isostatically pressed into pellets (10 mm diameter) for the electrochemical and ultrasonic measurements, while keeping a small fraction of powder for the diffraction studies.

X-Ray Powder Diffraction. X-ray diffraction measurements of $Li_{10}Ge_{1-x}Sn_xP_2S_{12}$ were carried out on a PANalytical Empyrean powder diffractometer in Bragg-Brentano $\theta - \theta$ geometry with Cu $K\alpha$ radiation ($\lambda_1 = 1.5405980\text{ \AA}$, $\lambda_2 = 1.5444260\text{ \AA}$). Measurements were carried out in the 2θ range between 10° and 90° with a step size of 0.026° . The counting time per step was 300 s. All powders were placed on (911)-oriented Silicon zero background holders and sealed using a 7.5 μm thick Kapton® polyimide film.

Rietveld-Analysis. Rietveld refinements were carried out using the TOPAS-Academic V6 software package (Bruker),³¹ using Thompson-Cox-Hastings pseudo-Voigt function for the profiles. The 2θ range of $10\text{--}26^\circ$ was excluded from the refinements in order to mitigate the interfering diffraction intensity from the polyimide film background. While appreciably better

fits were obtained with the abridged 2θ range, the exclusion of the lower 2θ region did not significantly impact the resultant structural data. As a starting point for the refinements, the structural model published by Weber *et al.* was used.²⁹ The atomic positions of lithium were taken from neutron powder diffraction results and were not refined. The atomic displacement parameters of all Li atoms were fixed to a B_{iso} of $7 \text{ \AA}^2$, as an average, extracted from the neutron diffraction data by Weber *et al.* on $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$.²⁹ For all other atoms, these parameters were refined. Additionally, Sn was allowed to occupy the Wyckoff $4d$ positions and constrained to the value of $g(\text{Ge}) + g(\text{Sn}) = 0.5$. Bond-lengths and polyhedral volumes were extracted from the Vesta software package (Version 3).³²

Electrochemical Impedance Spectroscopy. Temperature-dependent EIS was conducted to obtain the ionic conductivity and activation energy for each of the samples. Approximately 200 nm thick gold electrodes ($0.2 \text{ nm}\cdot\text{s}^{-1}$) were vapor deposited onto the pellets ($d = 8.2 \text{ mm}$) using a custom-built setup. Afterwards, the contacted pellets were sealed in pouch bags. Temperature-dependent EIS was then conducted. Conductivity measurements were carried out in the temperature range from 233 K to 333 K (climate chamber, Weiss Klimatechnik, 1 h equilibration time) using a Biologic SP300 impedance analyzer in the range of 7 MHz to 100 mHz with a sinusoidal amplitude of 10 mV. All fits were performed using the RelaxIS software package (rhd instruments, Version 3). The high frequency region ($< 2 \text{ MHz}$) was excluded from the fit because of high scattering stemming from the employed setup.

Ultrasonic Speed of Sound Measurements. An Epoch 600 (Olympus) with 5 MHz transducers was used on consolidated discs to measure the longitudinal and transverse speeds of sound. Since the samples are highly air sensitive and enclosing the pellets in pouches would prevent penetration of the transverse signals, the samples were coated with a thin layer ($< 200 \text{ nm}$) of gold in order to prevent side reactions between the couplant and the sample. All measurements were performed under a nitrogen atmosphere. The measurement uncertainty from the speed of sound data arises mainly from the uncertainty in the thickness and the porosity. As the measurement uncertainty for the thickness of the pellet is much higher than that of the gold layer, the thin layer of Au is not expected to add to the measurement uncertainty. In addition, no damping of the signal is expected, given the high crystallinity and high speed of sound in Au. This measurement procedure was already successfully employed to understand the lattice softness of $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$,²⁹ Li^+ -argyrodites⁴ and $\text{Na}_3\text{PS}_{4-x}\text{Se}_x$.³³ While in typical sound measurements on 100 % dense samples an uncertainty of 2 % can be

achieved, the measured values are systematically underestimated, as only 81 ± 1 % dense pellets were obtained. The uncertainties in the measurements were calculated using a linear error propagation of the predominant uncertainties (i.e. thickness of the pellet and standard deviation of the transit time of the sound waves). As the uncertainties do not reflect the influence of a scattering microstructure, the real uncertainties are expected to be slightly higher.^{4,33}

Ultrasonic measurements are a widely used and extremely accurate approach to determine the elastic properties and speeds of sound in materials. Such methods are a direct probe of the strength of the bonds themselves.^{34–37} Further, the obtained longitudinal and transverse speeds of sound (v_{long} and v_{trans}) can be used to calculate the mean speed of sound v_{mean} , as well as the Debye frequency ν_D via Equations 1–3, where V denotes the average volume per atom in the unit cell:^{38–40}

$$v_{\text{mean}}^3 = \frac{3}{v_{\text{long}}^{-3} + 2v_{\text{trans}}^{-3}} \quad \text{Eq. (1)}$$

$$\nu_D = \left(\frac{3}{4\pi V} \right)^{1/3} \cdot v_{\text{mean}} \quad \text{Eq. (2)}$$

These equations show that a decreasing speed of sound directly relates to a decreasing Debye frequency, which reflects the phonon vibrational frequencies of a material.

3. Results

Structural characterization. Figure 2a shows the X-ray diffraction patterns of all $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$ compounds with $x = 0, 0.33, 0.67$ and 1. An exemplary Rietveld refinement can be found in Figure 2b, showing the high quality of the structural characterization. The diffraction maxima within the isostructural $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$ patterns could be indexed to the tetragonal LGPS structure, crystallizing in the $P4_2/nmc$ space group (no. 137). Minor side phases, such as Li_2SnS_3 ⁴¹ for the Sn-compounds, were found at fractions below 3 wt% and are all provided in the Supporting Information. Additionally, all other diffraction patterns and corresponding Rietveld refinements, as well as the obtained structural data can also be found in the Supporting Information.

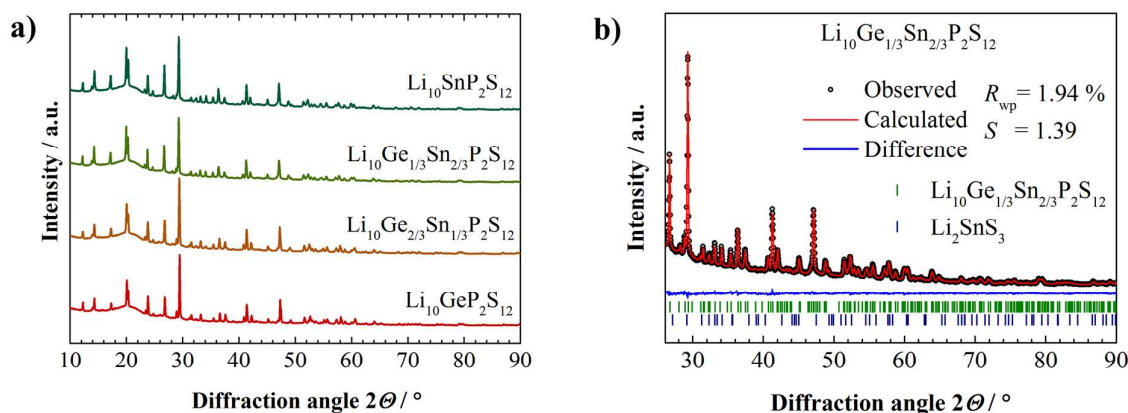


Figure 2: a) X-ray diffraction data of $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$. b) Exemplary Rietveld refinement of $\text{Li}_{10}\text{Ge}_{1/3}\text{Sn}_{2/3}\text{P}_2\text{S}_{12}$ with the goodness-of-fit S and profile residual R_{wp} . Experimental data are shown as points, the red line denotes the calculated pattern and the difference profile is shown in blue. Calculated positions of the Bragg reflections are shown as vertical ticks. A small fraction of the minor phase Li_2SnS_3 ⁴¹ can be seen and corresponds to 2.3 ± 0.5 wt %.

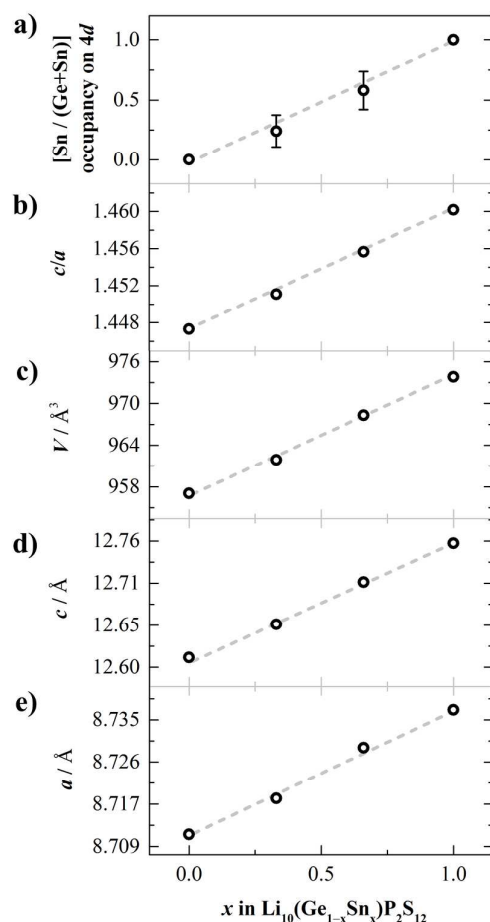


Figure 3: Dependence of the refined Sn occupancies, lattice parameters and lattice volume in $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$ on the nominal composition. a) Refined Sn occupancy on the Wyckoff 4d position. The error bars correspond to three times the statistical uncertainties (σ) obtained from the Rietveld refinements. It can be seen that the refinements reflect the nominal stoichiometry of the compound quite well. Experimentally, a small Sn-deficiency was found, which might be explained by the existence of the tin-containing side phases. b) c/a ratio and d)–e) Refined lattice parameters and lattice volume in $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$. A linear increase of the lattice parameters, the lattice volume and the c/a-ratio is observed, thereby obeying Vegard's law and indicating the successful synthesis of stable solid solutions.

The structural data obtained from the Rietveld refinements are shown in Figure 3. As expected (and previously shown by Kato *et al.*²³), the larger Sn^{4+} leads to a linear increase in the lattice parameters, the unit cell volume and the c/a-ratio of the tetragonal unit cell, all of which correspond well with Vegard's law. Moreover, the refinements confirm the increasing occupancy of Sn on the Wyckoff 4d Ge site, corroborating the successful synthesis of the

$\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$ solid solutions. In order to gain a more local understanding of the substitution-induced structural changes, the polyhedral volumes of the PS_4 and $(\text{M}/\text{P})\text{S}_4$ tetrahedra were examined (Figure 4a). As expected, the increasing fraction of Sn increases the volume of $(\text{M}/\text{P})\text{S}_4$, while the PS_4 tetrahedra retain their volume.^{26,42} Such variations in the polyhedral volumes prove the sole occupancy of Sn on the Wyckoff $4d$ position. In addition to the changes in the structural framework, the dependence of a representative Li-S polyhedral volume (i.e. $\text{Li}(2)\text{S}_6$) and the S3-S3 distance on the composition are shown in Figure 4b. With increasing unit cell volume and increasing c/a ratio, the Li-S polyhedra expand linearly, which directly represent the Li^+ - polyhedra in the x-y plane. A similar behavior was observed in the oxide garnet $\text{Li}_6\text{MLa}_2\text{Ta}_2\text{O}_{12}$ ⁴³ and indeed the broader diffusion pathway is expected to lead to a higher conductivity. However, the S3-S3 distance, corresponding to the apical sulfur of the Li1-S3 polyhedra, exhibit a decrease with increasing unit cell size and Sn-fraction. This decreasing distance likely restricts the Li1-Li1 jump along the tunnels in the z-direction, thereby acting as an additional bottleneck for lithium diffusion.

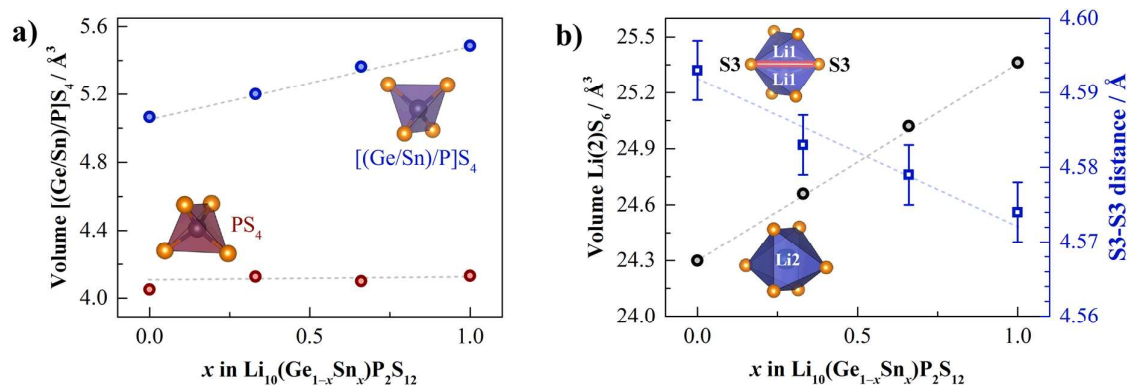


Figure 4: Effect of Sn-content on selected polyhedral volumes and the S3-S3 distance. a) While the $\text{P}(2b)$ -polyhedra do not show a change in volume, the $\text{M}(4d)$ -polyhedral volumes increase linearly, confirming the replacement of Ge^{4+} by Sn^{4+} . b) The Li-S-polyhedra expand with increasing unit cell volume, while the S3-S3 distance decreases, showing the dual effect of the alloying. The S3-S3 distance corresponds to the apical sulfur in the Li1-S polyhedra and a decrease in this distance likely acts as an additional bottleneck for diffusion.

Lattice dynamics of $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$. Beyond the structural changes, the underlying bonding interaction may also be altered upon moving from Ge^{4+} to the larger Sn^{4+} cation. Recently, a correlation was found between the softness of the anion lattice, *i.e.* the lattice dynamics, and the ionic transport in ionic conductors. Using solid solutions of $\text{Na}_3\text{PS}_{4-x}\text{Se}_x$ and $\text{Li}_6\text{PS}_5\text{X}$ ($\text{X} = \text{Cl}, \text{Br}, \text{I}$),^{4,33} it was possible to show that indeed a softer lattice lowers the activation barrier for ionic motion. Importantly, using speed of sound measurements, an average Debye frequency (see Equation 2) of the lattice can be obtained and employed as a metric for the softness of the lattice. Herein, ultrasonic speeds of sound were obtained for the solid solutions, *i.e.* the longitudinal v_{long} , transverse v_{trans} , and mean v_{mean} speeds of sound (Figure 5a). Due to the non-textured polycrystalline nature of the measured compounds, the obtained values reflect an average of the speeds of sound in the tetragonal crystal structure. With increasing Sn content, the speeds of sound in the $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$ series decrease, corresponding to a decrease of the Debye frequency ν_D (Figure 5b) and a softening of the lattice. These experimental results are in agreement with theoretical calculations, which reveal a higher bulk modulus and thus stiffer bonding properties in $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ relative to $\text{Li}_{10}\text{SnP}_2\text{S}_{12}$.⁴⁴

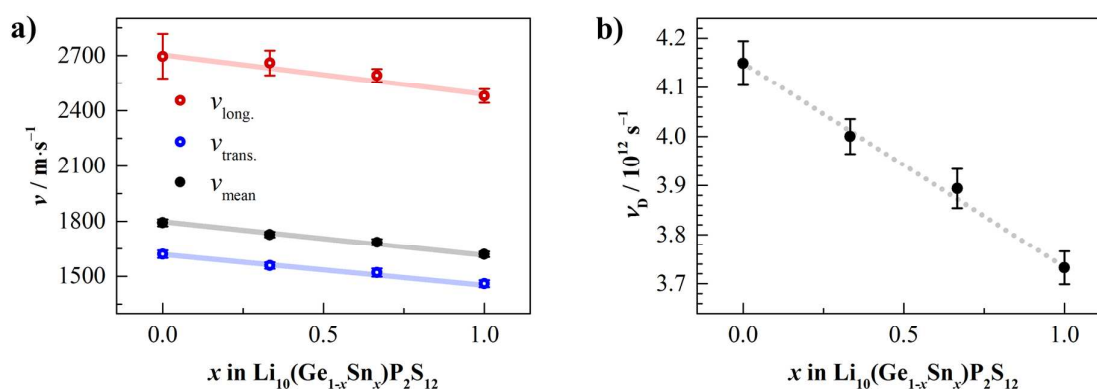


Figure 5: a) Longitudinal, transverse and mean speeds of sound in the $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$ series. The data show a decrease in the speeds of sound with increasing cation size within the anion framework. b) The higher lattice softness is further expressed by a decreasing Debye frequency.

Ionic transport. Temperature-dependent impedance spectroscopy was performed to assess the changes in the ionic conductivity upon Sn-substitution. Figure 5a shows the Arrhenius plots of the ionic conductivity for the $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$ solid solutions, while Figure 5b shows the impedance response at 233 K. The impedance data were fit with an equivalent

circuit consisting of one constant phase element (*CPE*) in parallel with a resistor (*R*), for the ionic transport, in series with a resistor that represents the blocking electrodes.

Within the applied temperature range, the angular frequency at the maximum of the semicircle $\omega_{\max}$ is outside the frequency window of the experiment. The capacitance *C* of the parallel combination *R/CPE* was therefore estimated using the Brug formula⁴⁵ ($C = [Q \cdot R^{1-\alpha}]^{1/\alpha}$; α = ideality factor). All geometrical capacitances were in the range of 37–49 pF·cm⁻² and therefore correspond well to bulk transport (see Table S7).⁴⁶ Additionally, the grain boundary capacitance of Li₁₀GeP₂S₁₂ was found to be in the range of 10 nF·cm⁻² and is not detected in the impedance spectra, indicating a negligible grain boundary resistance.²⁴ A decrease in the conductivity with increasing Sn fraction was observed, in good agreement with the literature.²³

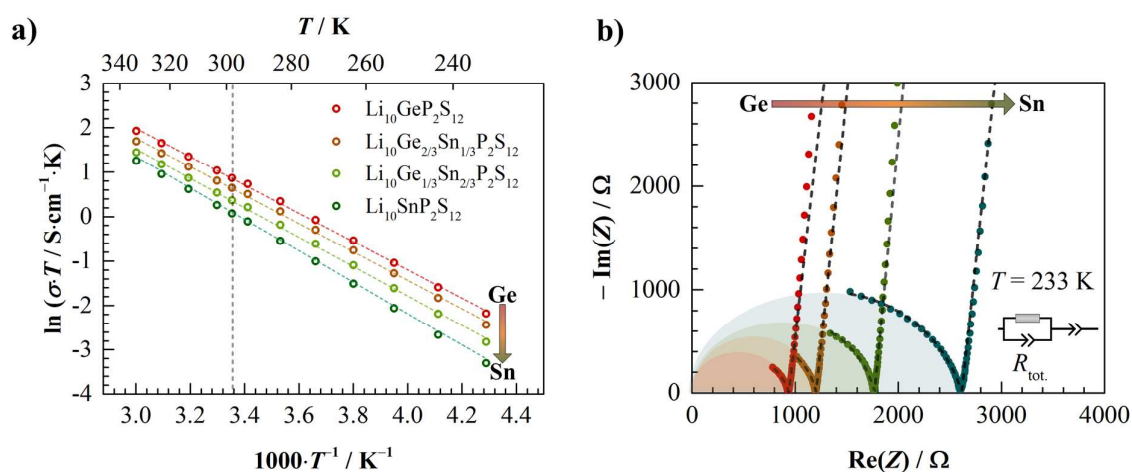


Figure 6: a) Arrhenius fits of the conductivity values for the Li₁₀Ge_xSn_{1-x}P₂S₁₂ series. The grey dashed line is a guide-to-the-eye corresponding to the room temperature measurements. b) Nyquist plots at 233 K showing the quality of the impedance data and the fit with the shown equivalent circuit.

4. Discussion

Upon moving from Ge⁴⁺ to Sn⁴⁺, the observed changes in the structure, ionic transport and lattice softness provide two likely explanations for the decreasing conductivity in Li₁₀Ge_{1-x}Sn_xP₂S₁₂. (1) The incorporation of larger Sn cations tightens the bottleneck for

lithium diffusion along the z-axis and (2) a softening of the (Ge/Sn)-S bonds with increasing Sn fraction causes stronger Li-S interactions.

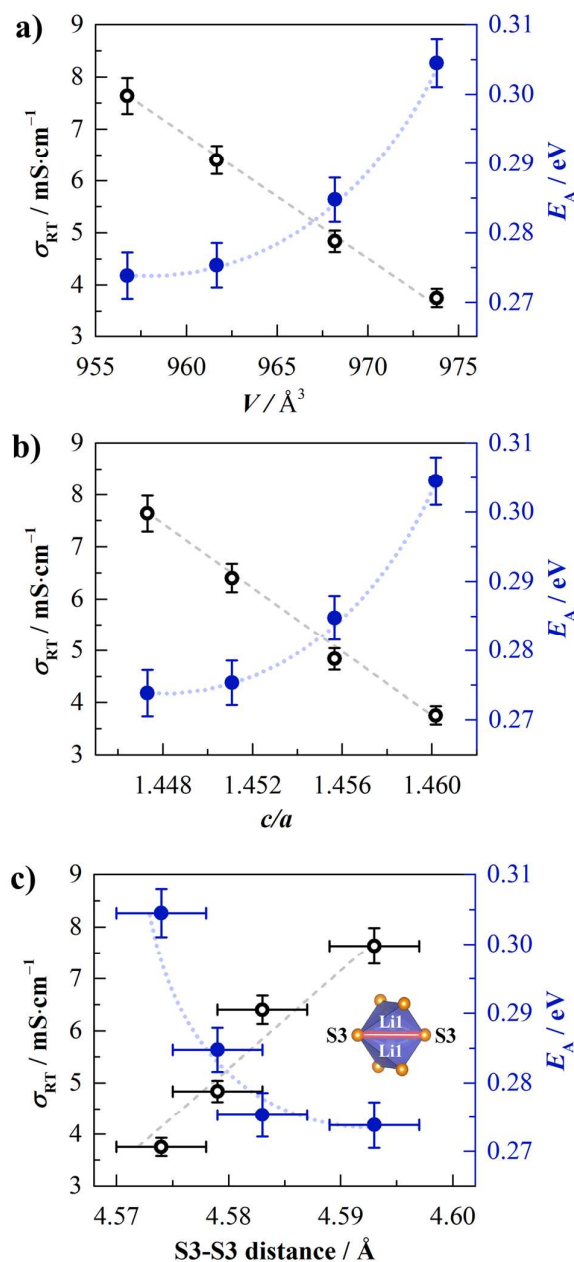


Figure 7: Interdependence of the structure and transport properties: ionic conductivity σ_{RT} and activation energy E_A on a) the volume V of the unit cell, b) the c/a ratio and c) the S3-S3 distance. The increases in the unit cell volume and the c/a ratio do not lead to an increase in the conductivity, but rather a decrease with a concurrent increase in the activation barrier. While a larger c/a ratio intuitively suggests a higher conductivity, in reality, the decreasing

S3-S3 distance acts as a bottleneck for lithium diffusion and restricts the motion along the z-axis.

In order to examine explanation (1) more closely, the interdependence between the structural properties and the ionic transport have been plotted in Figure 7. With increasing unit cell volume and c/a ratio, the activation barrier increases and the ionic conductivity decreases. As expected, the observed trends are in good agreement with previous experimental results from Kato et al.²³ and also correspond well with predicted theoretical trends drawn from ab initio molecular dynamics simulations from Ong et al.²¹ Interestingly, however, a decrease in the S3-S3 distance with increasing unit cell volume is noted, which acts as a bottleneck for ionic motion in the z-direction. The smaller distance forces the Li(1) to jump through a more narrow window, possibly destabilizing the transition state and raising the energetic barrier for ionic motion. It may also be that for the composition of $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, the interplay between possessing large enough polyhedra and large enough bottlenecks for ion jumps is optimal for transport. The data demonstrate that modulating the unit cell volume is often a good first approach toward influencing the conductivity in ionic conductors. However, more-local structural changes in less-isotropic materials can also counter the typical structural intuition. It can also be seen that the in-plane diffusion does not dominate the lithium mobility in this system, given that an increase in the c/a ratio does not result in an enhancement of the conductivity. Instead, these data suggest that the tunnels along the z-axis are indeed the dominating transport pathway in this structure, while the in-plane diffusion may only help to alleviate the known pitfalls of one-dimensional ionic conductors.

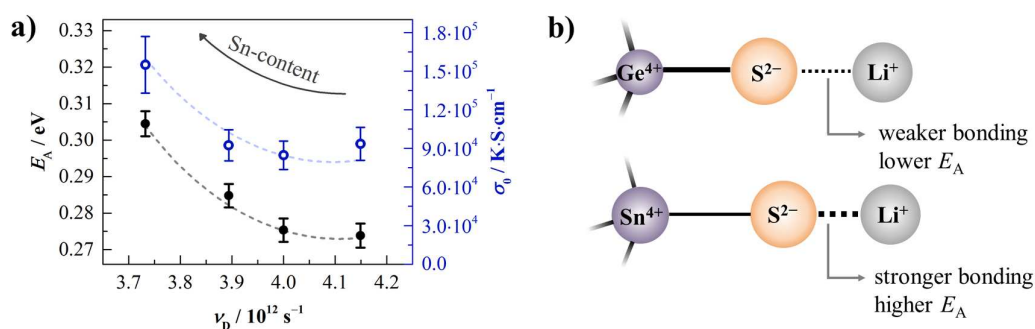


Figure 8: a) Correlation between ionic transport and lattice dynamics, represented by the Debye frequency ν_D . The activation energy E_A and the prefactor σ_0 show an increase with increasing overall lattice softness. b) Sketch showing the changing Coulombic interactions between sulfur and lithium resulting from the different M-S- bonding characteristics.

In addition to the structural effects on the conductivity, the influence of the chemical environment surrounding the Li atoms must also be considered. Figure 8a shows the influence of the lattice softness on the ionic transport. With increasing lattice softness, *i.e.* decreasing Debye frequency ν_D , the activation barrier and the Arrhenius pre-factor are found to increase. At first glance, the obtained correlation between the lattice dynamics and the transport appear to contradict the results found for $\text{Na}_3\text{PS}_{4-x}\text{Se}_x$ and $\text{Li}_6\text{PS}_5\text{X}$ ($\text{X} = \text{Cl}, \text{Br}, \text{I}$), as the conductivity is decreasing with increasing lattice softness.^{4,33} However, in the two model systems, the full polarizability of the anion framework was varied and with it, not only the lattice softness, but all cation-anion interactions. Herein, the substitution of Ge with Sn within the (Ge/Sn) S_4 polyhedra makes the $\text{M}^{4+} - \text{S}^{2-}$ bonds softer and more elongated (see Figure 8b). This variation in the bonding interactions is reflected in the softening of the lattice, as the Debye frequency corresponds to an averaging of the bonding characteristics over the whole structure. However, for ionic transport, the local bonding interactions (*i.e.* the direct chemical neighborhood of $\text{Li}^+ - \text{S}^{2-}$) are most important. Thus, the longer $\text{Sn}^{4+} - \text{S}^{2-}$ bonds and higher electronegativity of Sn vs. Ge (1.7 vs. 2.0, Allred-Rochow scale of electronegativity)⁴⁷ lead to more electron density on the S^{2-} atoms, which in turn may lead to stronger $\text{Li}^+ - \text{S}^{2-}$ Coulombic attractions, thereby inhibiting ionic transport. The stronger Coulombic attractions lead to a higher activation barrier for ionic motion and with it a higher pre-factor (see Meyer-Neldel plot in the Supporting Information). It should also be noted that, similar inductive effects⁴⁸ have been found in $\text{Na}_3\text{P}_{1-x}\text{As}_x\text{S}_4$,⁴⁹ in which the longer and weaker $\text{As}^{5+} - \text{S}^{2-}$ bonds led to stronger $\text{Na}^+ - \text{S}^{2-}$ interactions and a lower conductivity.

The observed correlations between the local structural environment surrounding lithium and the associated bonding interactions provide plausible explanations for the observed transport behavior in $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$. The data suggest that both stronger Li-S interactions and a decreasing bottleneck size are affecting the transition state for lithium diffusion, and that the stronger Coulombic attractions may in fact stem from the shorter S3-S3 distance, thereby decreasing the ionic conductivity.

5. Conclusion

In this work, the influence of a changing structure and local bonding interactions was investigated in a series of $\text{Li}_{10}\text{Ge}_{1-x}\text{Si}_x\text{P}_2\text{S}_{12}$ solid solutions. Using a combination of X-ray

diffraction, speed of sound measurements and impedance spectroscopy, we obtain likely explanations for the decreased ionic conductivity upon Sn-substitution, despite the observed lattice expansion. First, this work shows that a structural bottleneck along the diffusion channels in the z -direction tightens during the Sn-substitution and concurrent unit cell expansion. Second, the influence of the lattice softness suggests that Coulombic bonding interactions between Li^+ and S^{2-} become stronger, which further increases the activation barrier. The data suggest that there is more to the underlying lithium diffusion mechanism in $\text{Li}_{10}\text{MP}_2\text{S}_{12}$ and that the tunnels, rather than the in-plane pathways, may indeed dominate the ionic transport. This work provides a better understanding of structure-property relationships in ionic conductors and highlights the importance of investigating the structure from a more local standpoint.

Supporting Information

All structural data as obtained from Rietveld refinements against X-ray powder diffraction data between $26\text{--}90^\circ$ 2θ are tabulated here. Fits of the entire $10\text{--}90^\circ$ 2θ range are also provided. Additionally, a Meyer-Neldel plot and a literature survey of lattice parameters and ionic conductivities for $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ and $\text{Li}_{10}\text{SnP}_2\text{S}_{12}$ can be found here. The SI also contains the measured lattice dynamical values and transport data for all synthesized compounds. Finally, the resultant .cif files of the crystallographic data can be found here.

AUTHOR INFORMATION

Corresponding Authors

*wolfgang.g.zeier@phys.chemie.uni-giessen.de;

Notes

The authors declare no competing financial interests.

Acknowledgements

The research was supported by the Deutsche Forschungsgemeinschaft (DFG) under grant number ZE 1010/4-1. S.C. gratefully acknowledges the Alexander von Humboldt Foundation

for financial support through a Postdoctoral Fellowship.

References

- (1) Janek, J.; Zeier, W. G. A solid future for battery development. *Nat. Energy* **2016**, *1*, 16141.
- (2) Liu, D.; Zhu, W.; Feng, Z.; Guerfi, A.; Vijh, A.; Zaghib, K. Recent progress in sulfide-based solid electrolytes for Li-ion batteries. *Materials Science and Engineering: B* **2016**, *213*, 169–176.
- (3) Bachman, J. C.; Muy, S.; Grimaud, A.; Chang, H.-H.; Pour, N.; Lux, S. F.; Paschos, O.; Maglia, F.; Lupart, S.; Lamp, P. Inorganic solid-state electrolytes for lithium batteries: mechanisms and properties governing ion conduction. *Chem. Rev.* **2015**, *116*, 140–162.
- (4) Kraft, M. A.; Culver, S. P.; Calderon, M.; Böcher, F.; Krauskopf, T.; Senyshyn, A.; Dietrich, C.; Zevalkink, A.; Janek, J.; Zeier, W. G. Influence of Lattice Polarizability on the Ionic Conductivity in the Lithium Superionic Argyrodites $\text{Li}_6\text{PS}_5\text{X}$ ($\text{X} = \text{Cl}, \text{Br}, \text{I}$). *J. Am. Chem. Soc.* **2017**, *139*, 10909–10918.
- (5) Kong, S.-T. Synthesis, Characterization and Structural Chemistry of Lithium Argyrodites. Dissertation, Universität Siegen, Siegen, 2010.
- (6) Deiseroth, H.; Maier, J.; Weichert, K.; Nickel, V.; Kong, S.; Reiner, C. Li_7PS_6 and $\text{Li}_6\text{PS}_5\text{X}$ ($\text{X} = \text{Cl}, \text{Br}, \text{I}$): Possible Three-dimensional Diffusion Pathways for Lithium Ions and Temperature Dependence of the Ionic Conductivity by Impedance Measurements. *Z. anorg. allg. Chem.* **2011**, *637*, 1287–1294.
- (7) Deiseroth, H.; Kong, S.; Eckert, H.; Vannahme, J.; Reiner, C.; Zaiß, T.; Schlosser, M. $\text{Li}_6\text{PS}_5\text{X}$: A Class of Crystalline Li-Rich Solids With an Unusually High Li^+ Mobility. *Angew. Chem. Int. Edit.* **2008**, *47*, 755–758.
- (8) Minafra, N.; Culver, S. P.; Krauskopf, T.; Senyshyn, A.; Zeier, W. G. Effect of Si substitution on the structural and transport properties of superionic Li-argyrodites. *J. Mater. Chem. A* **2018**, *6*, 645–651.
- (9) Dietrich, C.; Sadowski, M.; Siculo, S.; Weber, D. A.; Sedlmaier, S. J.; Weldert, K. S.; Indris, S.; Albe, K.; Janek, J.; Zeier, W. G. Local Structural Investigations, Defect Formation, and Ionic Conductivity of the Lithium Ionic Conductor $\text{Li}_4\text{P}_2\text{S}_6$. *Chem. Mater.* **2016**, *28*, 8764–8773.

(10) Dietrich, C.; Weber, D. A.; Culver, S.; Senyshyn, A.; Sedlmaier, S. J.; Indris, S.; Janek, J.; Zeier, W. G. Synthesis, Structural Characterization, and Lithium Ion Conductivity of the Lithium Thiophosphate $\text{Li}_2\text{P}_2\text{S}_6$. *Inorg. Chem.* **2017**.

(11) Dietrich, C.; Weber, D. A.; Sedlmaier, S. J.; Indris, S.; Culver, S. P.; Walter, D.; Janek, J.; Zeier, W. G. Lithium ion conductivity in Li_2S – P_2S_5 glasses—building units and local structure evolution during the crystallization of superionic conductors Li_3PS_4 , $\text{Li}_7\text{P}_3\text{S}_{11}$ and $\text{Li}_4\text{P}_2\text{S}_7$. *J. Mater. Chem. A* **2017**, *5*, 18111–18119.

(12) Tatsumisago, M.; Hayashi, A. Sulfide Glass-Ceramic Electrolytes for All-Solid-State Lithium and Sodium Batteries. *Int. J. Appl. Glas. Sci.* **2014**, *5*, 226–235.

(13) Seino, Y.; Ota, T.; Takada, K.; Hayashi, A.; Tatsumisago, M. A sulphide lithium super ion conductor is superior to liquid ion conductors for use in rechargeable batteries. *Energy Environ. Sci.* **2014**, *7*, 627–631.

(14) Minami, T.; Hayashi, A.; Tatsumisago, M. Recent progress of glass and glass-ceramics as solid electrolytes for lithium secondary batteries. *Solid State Ionics* **2006**, *177*, 2715–2720.

(15) Ohara, K.; Mitsui, A.; Mori, M.; Onodera, Y.; Shiotani, S.; Koyama, Y.; Orikasa, Y.; Murakami, M.; Shimoda, K.; Mori, K. Structural and electronic features of binary Li_2S – P_2S_5 glasses. *Sci. Rep.* **2016**, *6*.

(16) Kamaya, N.; Homma, K.; Yamakawa, Y.; Hirayama, M.; Kanno, R.; Yonemura, M.; Kamiyama, T.; Kato, Y.; Hama, S.; Kawamoto, K. A lithium superionic conductor. *Nat. Mater.* **2011**, *10*, 682–686.

(17) Kato, Y.; Hori, S.; Saito, T.; Suzuki, K.; Hirayama, M.; Mitsui, A.; Yonemura, M.; Iba, H.; Kanno, R. High-power all-solid-state batteries using sulfide superionic conductors. *Nat. Energy* **2016**, *1*, 16030.

(18) Zhang, W.; Weber, D. A.; Weigand, H.; Arlt, T.; Manke, I.; Schröder, D.; Koerver, R.; Leichtweiss, T.; Hartmann, P.; Zeier, W. G. *et al.* Interfacial Processes and Influence of Composite Cathode Microstructure Controlling the Performance of All-Solid-State Lithium Batteries. *ACS Appl. Mater. Interfaces* **2017**, *9*, 17835–17845.

(19) Zhang, W.; Schröder, D.; Arlt, T.; Manke, I.; Koerver, R.; Pinedo, R.; Weber, D. A.; Sann, J.; Zeier, W. G.; Janek, J. (Electro) chemical expansion during cycling: monitoring the pressure changes in operating solid-state lithium batteries. *J. Mater. Chem. A* **2017**, *5*, 9929–9936.

- (20) Bron, P.; Johansson, S.; Zick, K.; Schmedt auf der G nne, J.; Dehnen, S.; Roling, B. $\text{Li}_{10}\text{SnP}_2\text{S}_{12}$: an affordable lithium superionic conductor. *J. Am. Chem. Soc.* **2013**, *135*, 15694–15697.
- (21) Ong, S. P.; Mo, Y.; Richards, W. D.; Miara, L.; Lee, H. S.; Ceder, G. Phase stability, electrochemical stability and ionic conductivity of the $\text{Li}_{10\pm 1}\text{MP}_2\text{X}_{12}$ (M = Ge, Si, Sn, Al or P, and X = O, S or Se) family of superionic conductors. *Energ. Environ. Sci.* **2013**, *6*, 148–156.
- (22) Hori, S.; Suzuki, K.; Hirayama, M.; Kato, Y.; Saito, T.; Yonemura, M.; Kanno, R. Synthesis, structure, and ionic conductivity of solid solution, $\text{Li}_{10+\delta}\text{M}_{1+\delta}\text{P}_{2-\delta}\text{S}_{12}$ (M = Si, Sn). *Faraday Discuss.* **2015**, *176*, 83–94.
- (23) Kato, Y.; Saito, R.; Sakano, M.; Mitsui, A.; Hirayama, M.; Kanno, R. Synthesis, structure and lithium ionic conductivity of solid solutions of $\text{Li}_{10}(\text{Ge}_{1-x}\text{M}_x)\text{P}_2\text{S}_{12}$ (M = Si, Sn). *J. Power Sources* **2014**, *271*, 60–64.
- (24) Bron, P.; Dehnen, S.; Roling, B. $\text{Li}_{10}\text{Si}_{0.3}\text{Sn}_{0.7}\text{P}_2\text{S}_{12}$ —A low-cost and low-grain-boundary-resistance lithium superionic conductor. *J. Power Sources* **2016**, *329*, 530–535.
- (25) Kuhn, A.; Gerbig, O.; Zhu, C.; Falkenberg, F.; Maier, J.; Lotsch, B. V. A new ultrafast superionic Li-conductor: ion dynamics in $\text{Li}_{11}\text{Si}_2\text{PS}_{12}$ and comparison with other tetragonal LGPS-type electrolytes. *Phys. Chem. Chem. Phys.* **2014**, *16*, 14669–14674.
- (26) Sun, Y.; Suzuki, K.; Hori, S.; Hirayama, M.; Kanno, R. Superionic Conductors: $\text{Li}_{10+\delta}[\text{Sn}_y\text{Si}_{1-y}]_{1+\delta}\text{P}_{2-\delta}\text{S}_{12}$ with a $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ -type Structure in the Li_3PS_4 – Li_4SnS_4 – Li_4SiS_4 Quasi-ternary System. *Chem. Mater.* **2017**, *29*, 5858–5864.
- (27) Kwon, O.; Hirayama, M.; Suzuki, K.; Kato, Y.; Saito, T.; Yonemura, M.; Kamiyama, T.; Kanno, R. Synthesis, structure, and conduction mechanism of the lithium superionic conductor $\text{Li}_{10+\delta}\text{Ge}_{1+\delta}\text{P}_{2-\delta}\text{S}_{12}$. *J. Mater. Chem. A* **2015**, *3*, 438–446.
- (28) Mo, Y.; Ong, S. P.; Ceder, G. First principles study of the $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ lithium super ionic conductor material. *Chem. Mater.* **2012**, *24*, 15–17.
- (29) Weber, D. A.; Senyshyn, A.; Weldert, K. S.; Wenzel, S.; Zhang, W.; Kaiser, R.; Berendts, S.; Janek, J.; Zeier, W. G. Structural insights and 3D diffusion pathways within the lithium superionic conductor $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$. *Chem. Mater.* **2016**, *28*, 5905–5915.
- (30) Liang, X.; Wang, L.; Jiang, Y.; Wang, J.; Luo, H.; Liu, C.; Feng, J. In-channel and In-plane Li ion diffusions in the superionic conductor $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ Probed by solid-state NMR. *Chem. Mater.* **2015**, *27*, 5503–5510.
- (31) Coelho, A. A. Topas-Academic. *Coelho Software, Brisbane, Australia* **2007**.

- (32) Momma, K.; Izumi, F. VESTA 3 for three-dimensional visualization of crystal, volumetric and morphology data. *J. Appl. Chem.* **2011**, *44*, 1272–1276.
- (33) Krauskopf, T.; Pompe, C.; Kraft, M. A.; Zeier, W. G. Influence of Lattice Dynamics on Na^+ Transport in the Solid Electrolyte $\text{Na}_3\text{PS}_{4-x}\text{Se}_x$. *Chem. Mater.* **2017**, *29*, 8859–8869.
- (34) Zeier, W. G.; Zevalkink, A.; Gibbs, Z. M.; Hautier, G.; Kanatzidis, M. G.; Snyder, G. J. Thinking like a chemist: intuition in thermoelectric materials. *Angew. Chem. Int. Edit.* **2016**, *55*, 6826–6841.
- (35) Heinrich, C. P.; Day, T. W.; Zeier, W. G.; Snyder, G. J.; Tremel, W. Effect of Isovalent Substitution on the Thermoelectric Properties of the $\text{Cu}_2\text{ZnGeSe}_{4-x}\text{S}_x$ Series of Solid Solutions. *J. Am. Chem. Soc.* **2013**, *136*, 442–448.
- (36) Weldert, K. S.; Zeier, W. G.; Day, T. W.; Panthöfer, M.; Snyder, G. J.; Tremel, W. Thermoelectric transport in Cu_7PSe_6 with high copper ionic mobility. *J. Am. Chem. Soc.* **2014**, *136*, 12035–12040.
- (37) Toberer, E. S.; Zevalkink, A.; Snyder, G. J. Phonon engineering through crystal chemistry. *J. Mater. Sci.* **2011**, *21*, 15843–15852.
- (38) Anderson, O. L. A simplified method for calculating the Debye temperature from elastic constants. *J. Phys. Chem. Solids* **1963**, *24*, 909–917.
- (39) Zevalkink, A.; Toberer, E. S.; Zeier, W. G.; Flage-Larsen, E.; Snyder, G. J. Ca_3AlSb_3 : an inexpensive, non-toxic thermoelectric material for waste heat recovery. *Energ. Environ. Sci.* **2011**, *4*, 510–518.
- (40) May, A. F.; Snyder, G. J. Introduction to Modeling Thermoelectric Transport at High Temperatures. In *Thermoelectrics Handbook: Thermoelectrics and Its Energy Harvesting*; Rowe, D. M., Ed.; CRC Press: Boca Raton, FL, 2012.
- (41) Brant, J. A.; Massi, D. M.; Holzwarth, N. A.; MacNeil, J. H.; Douvalis, A. P.; Bakas, T.; Martin, S. W.; Gross, M. D.; Aitken, J. A. Fast Lithium Ion Conduction in Li_2SnS_3 : Synthesis, Physicochemical Characterization, and Electronic Structure. *Chem. Mater.* **2014**, *27*, 189–196.
- (42) Hori, S.; Suzuki, K.; Hirayama, M.; Kato, Y.; Saito, T.; Yonemura, M.; Kanno, R. Synthesis, structure, and ionic conductivity of solid solution, $\text{Li}_{10+\delta}\text{M}_{1+\delta}\text{P}_{2-\delta}\text{S}_{12}$ (M= Si, Sn). *Faraday Discuss.* **2014**, *176*, 83–94.
- (43) Zeier, W. G.; Zhou, S.; Lopez-Bermudez, B.; Page, K.; Melot, B. C. Dependence of the Li-ion conductivity and activation energies on the crystal structure and ionic radii in $\text{Li}_6\text{MLa}_2\text{Ta}_2\text{O}_{12}$. *ACS Appl. Mater. Interfaces* **2014**, *6*, 10900–10907.

- (44) Deng, Z.; Wang, Z.; Chu, I.-H.; Luo, J.; Ong, S. P. Elastic properties of alkali superionic conductor electrolytes from first principles calculations. *J. Electrochem. Soc.* **2016**, *163*, A67-A74.
- (45) Brug, G. J.; van den Eeden, A.; Sluyters-Rehbach, M.; Sluyters, J. H. The analysis of electrode impedances complicated by the presence of a constant phase element. *J. Electroanal. Chem.* **1984**, *176*, 275–295.
- (46) Irvine, J. T. S.; Sinclair, D. C.; West, A. R. Electroceramics: characterization by impedance spectroscopy. *Adv. Mater.* **1990**, *2*, 132–138.
- (47) Allred, A. L.; Rochow, E. G. A scale of electronegativity based on electrostatic force. *J. Inorg. Nuclear Chem.* **1958**, *5*, 264–268.
- (48) Etourneau, J.; Portier, J.; Menil, F. The role of the inductive effect in solid state chemistry: how the chemist can use it to modify both the structural and the physical properties of the materials. *Journal of Alloys and Compounds* **1992**, *188*, 1–7.
- (49) Yu, Z.; Shang, S.; Seo, J.; Wang, D.; Luo, X.; Huang, Q.; Chen, S.; Lu, J.; Li, X.; Liu, Z. Exceptionally High Ionic Conductivity in $\text{Na}_3\text{P}_{0.62}\text{As}_{0.38}\text{S}_4$ with Improved Moisture Stability for Solid-State Sodium-Ion Batteries. *Adv. Mater.* **2017**, *29*, 1605561.

Table of contents image

